

SHOWA STATION, Antarctica

WMO: 895320

Lat: 69.005S

Lon: 39.581E

Elev: 60

StdP: 14.66

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-28.1	-23.4	-35.4	0.7	-26.2	-32.0	0.9	-21.9	67.9	14.4	62.0	13.7	5.4	200	1.243

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.7	38.3	32.3	36.6	31.3	35.1	30.5	33.1	36.9	32.0	35.3	31.1	34.3	8.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
30.3	24.5	32.7	28.9	22.9	32.2	27.6	21.5	31.8	12.3	37.2	11.9	35.5	11.5	34.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 60.6	52.0	44.8	DB	-32.2	42.1	5.6	2.0	-36.2	43.5	-39.5	44.6	-42.6	45.7	-46.7	47.1	(4)
(5)			WB	-32.4	35.1	5.6	1.4	-36.4	36.1	-39.7	36.9	-42.8	37.7	-46.8	38.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	30.0	26.6	19.6	13.3	7.2	4.5	0.5	-1.7	-1.1	7.7	20.3	28.9
	DBStd	13.96	3.32	4.36	6.00	7.93	9.38	10.02	11.23	10.94	10.71	8.50	6.59	3.47
	HDD50	13537	620	655	943	1101	1325	1365	1535	1602	1534	1313	891	653
	HDD65	19012	1085	1075	1408	1551	1790	1815	2000	2067	1984	1778	1341	1118
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	15.0	11.0	15.6	17.3	19.0	17.2	16.5	15.8	14.3	13.9	14.5	13.7	12.0
Precipitation	PrecAvg	16.4	1.4	1.8	1.9	1.8	1.5	1.3	1.3	1.1	0.8	1.2	1.2	1.1
	PrecMax	21.2	2.6	3.2	2.8	2.9	2.9	2.5	3.1	2.1	1.8	2.4	2.5	2.4
	PrecMin	12.5	0.6	1.0	0.6	0.5	0.5	0.3	0.5	0.4	0.2	0.5	0.1	0.2
	PrecStd	2.0	0.6	0.6	0.5	0.5	0.7	0.6	0.6	0.4	0.4	0.4	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.2	38.5	31.9	29.7	26.2	26.0	23.0	22.3	22.9	27.0	35.6	40.3
		MCWB	34.0	32.2	29.3	29.1	25.0	24.9	21.9	20.9	21.3	25.4	29.7	33.7
	2%	DB	38.8	35.5	30.4	26.7	24.2	23.9	20.5	19.1	19.8	24.9	33.2	37.8
		MCWB	32.5	30.8	28.1	25.5	22.8	22.7	19.2	17.8	18.4	23.4	28.4	32.2
	5%	DB	37.1	33.8	28.8	24.9	22.2	21.6	18.0	16.6	16.6	22.1	30.9	36.2
		MCWB	31.7	30.3	26.5	23.5	20.9	20.3	16.7	15.3	15.2	20.5	26.8	31.2
	10%	DB	35.7	32.5	27.3	23.4	19.5	19.0	15.8	13.9	13.6	19.9	29.0	34.7
		MCWB	30.8	29.5	24.9	21.8	18.2	17.8	14.4	12.6	12.0	18.2	25.4	30.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.7	33.6	30.3	29.2	25.8	25.4	22.3	21.5	21.8	26.1	30.5	35.0
		MCDB	39.8	36.5	31.1	29.5	26.0	25.8	22.7	22.3	22.5	26.5	34.3	39.3
	2%	WB	33.2	32.2	28.7	25.9	23.1	22.8	19.4	18.0	18.5	23.4	28.8	32.8
		MCDB	37.4	34.0	29.9	26.5	23.8	23.5	20.4	18.9	19.8	24.5	32.5	36.8
	5%	WB	32.2	31.2	26.8	23.8	20.9	20.4	16.8	15.5	15.5	20.6	27.2	31.5
		MCDB	36.1	33.2	28.5	24.7	22.1	21.5	17.9	16.5	16.6	21.9	30.5	35.3
	10%	WB	31.3	30.0	25.2	22.0	18.3	17.9	14.4	12.5	12.1	18.2	25.7	30.5
		MCDB	34.9	32.4	27.0	23.1	19.5	19.1	15.7	13.9	13.5	19.8	28.8	34.1
Mean Daily Temperature Range	5% DB	MDBR	7.7	7.0	7.4	8.1	8.7	9.9	10.1	11.0	10.7	9.6	9.5	7.9
		MCDBR	9.3	7.1	6.2	6.1	7.3	9.6	10.0	11.4	11.3	9.1	10.2	9.5
		MCWBR	6.5	5.7	6.3	6.6	7.4	9.6	10.1	11.1	11.1	8.6	8.4	7.0
	5% WB	MCDBR	8.0	5.8	6.2	6.3	7.0	9.6	10.0	11.3	11.3	8.6	9.1	8.5
		MCWBR	6.0	5.1	6.6	6.9	7.2	9.6	10.2	11.1	11.3	8.3	7.7	6.6
Clear-Sky Solar Irradiance	taub	0.257	0.252	0.230	0.185	0.129	N/A	0.128	0.168	0.201	0.222	0.238	0.257	
	taud	2.267	2.292	2.312	2.138	1.911	N/A	1.892	2.093	2.251	2.268	2.285	2.254	
	Ebn at Noon	314	296	268	196	34	0	26	201	277	307	321	320	
	Edh at Noon	36	30	21	11	1	0	1	11	22	30	35	37	
All-Sky Solar Radiation	RadAvg	2473	1584	777	206	19	0	3	114	579	1347	2301	2821	
	RadStd	115	89	50	17	2	0	0	7	30	51	90	92	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page